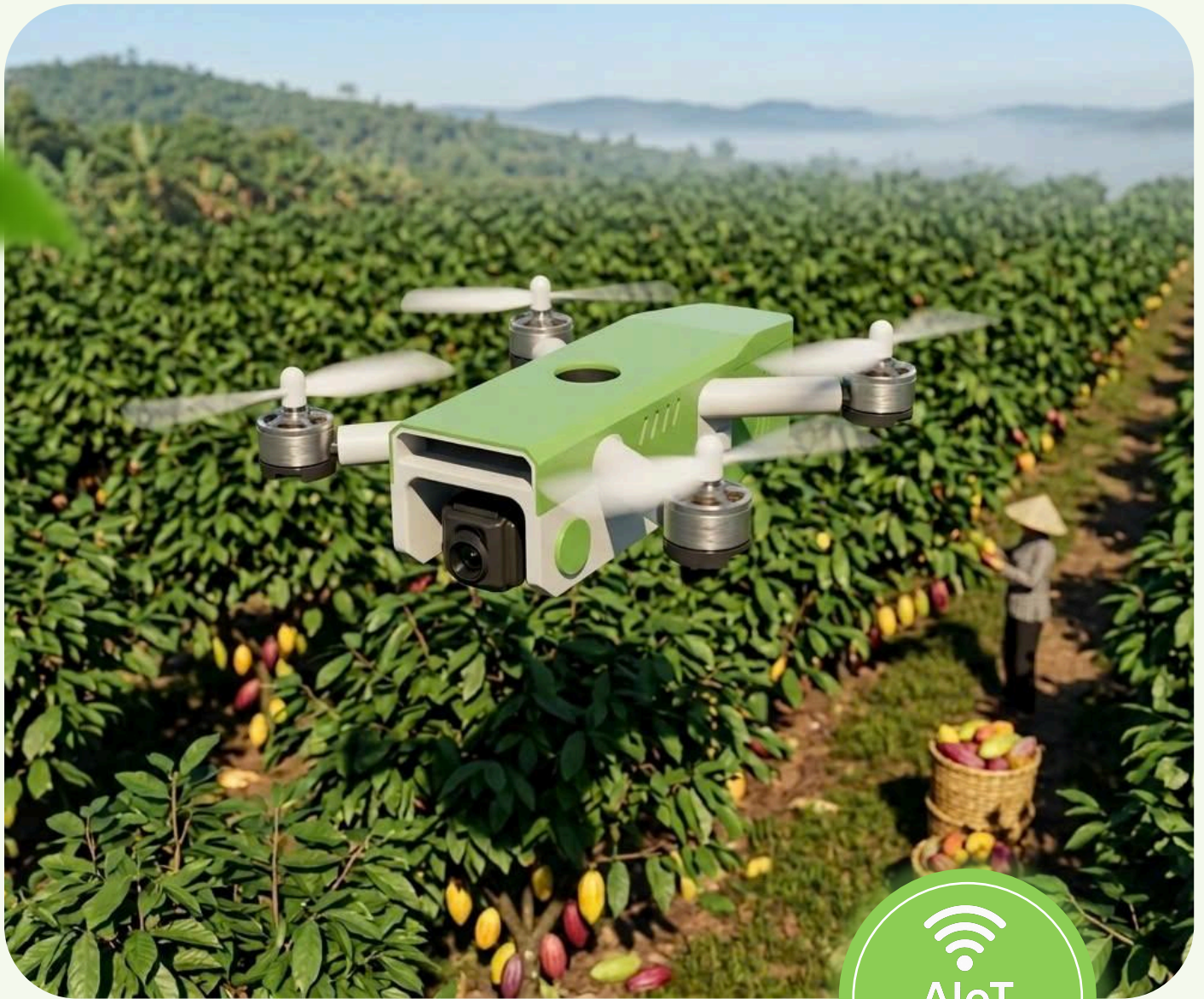




AIoT PLATFORM FOR NEXT-GEN FARMING

Smart Sensors

Accurate Data

Sustainable Growth

BUILDING A TRUSTED DATA LAYER FOR SUSTAINABLE FARMING

At AgriBeacon, we believe modern agriculture should combine practical experience with the power of technology. Every farming decision should rely on accurate data, captured from the field and verifiable throughout operations.



Continuous Field Data

Signals from soil, water, crops, IoT devices, drones, GPS, and daily operations are captured in real time, creating a reliable data layer for modern farm management.



AI-Powered Action

AI goes beyond data analysis by turning field signals into alerts, recommendations, and specific tasks, helping operators make faster decisions.



Transparent Value Chain

Operational data is standardized into verifiable records, helping farms improve management capabilities and enabling buyers and exporters to build market trust.

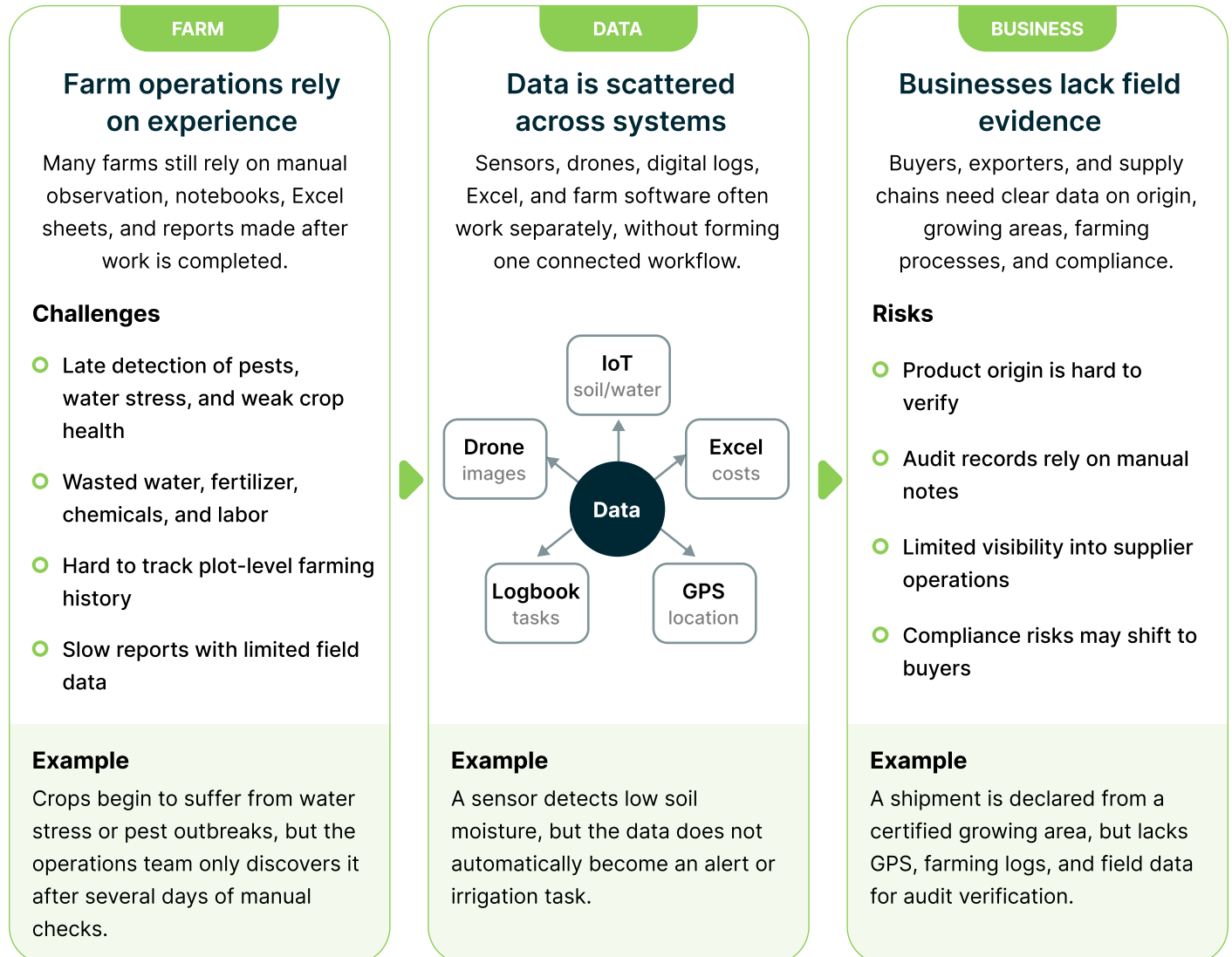


With AgriBeacon, data becomes the foundation for improving productivity, controlling costs, managing risks, and proving product value across the supply chain.



AGRICULTURE HAS MANY TOOLS, BUT DATA REMAINS FRAGMENTED

Farms, cooperatives, and agribusinesses are facing growing pressure around productivity, costs, labor, traceability, and compliance standards.



CORE BOTTLENECK

What is missing is not another separate tool, but a unified operating layer that connects source data, field activities, people, devices, AI, and verified records.

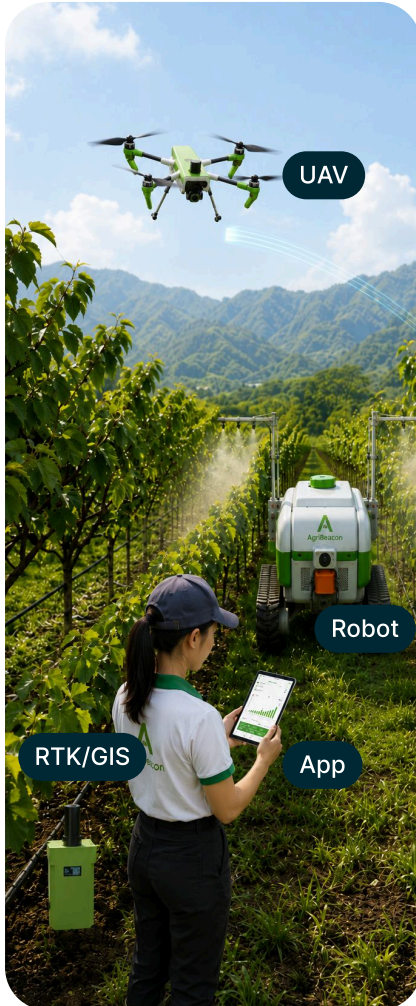
When data is captured at the right time, in the right place, and linked to the right activity, farms can make faster decisions, businesses can verify origin more clearly, and the entire value chain becomes more transparent.

AGRIBEACON HIGH-TECH ECOSYSTEM

AgriBeacon connects field data collection, farm operations, AI intelligence, and traceability into one unified workflow.

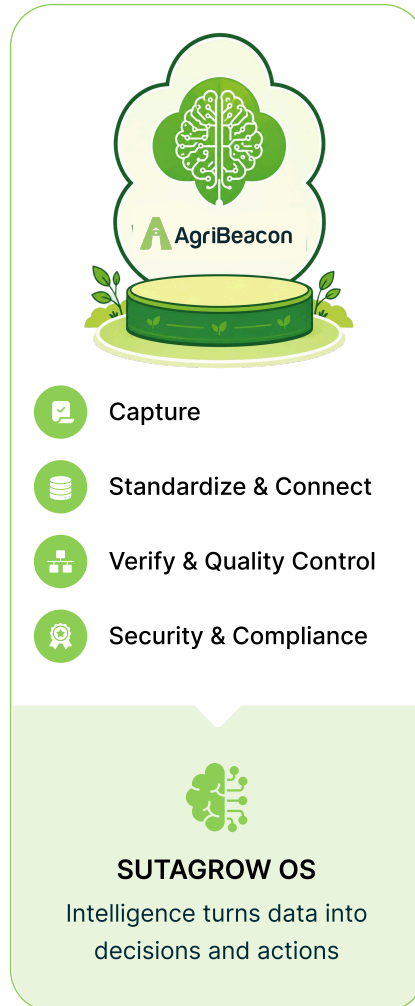
1

Field



2

AI Brain & Cloud



3

Commercial Connection



Farm Operations

- Protect productivity
- Optimize resources
- Improve labor efficiency
- Support sustainable practices
- Real-time visibility



Trusted Evidence for Buyers

- Farm origin & identity
- Input & activity records
- Compliance & standards
- End-to-end traceability
- Audit-ready evidence

Value Created



Higher Productivity

Better year after year



Optimized Inputs

Lower costs, less waste



Sustainable Farming

Protecting soil, nurturing future generations



Stronger Relationships

Built on trust & transparency

RTK ROVER/BASE: MONRTK

RTK-based crop and farm mapping solution



MONRTK applies high-precision RTK positioning technology to map farms, plot boundaries, crop locations, and movement routes for agricultural machinery.



High-Precision Positioning

Locate positions with centimeter-level accuracy.



Detailed Mapping

Map boundaries, crops, routes, and farm infrastructure.



Time Saving

Survey faster and reduce reliance on manual measurement.

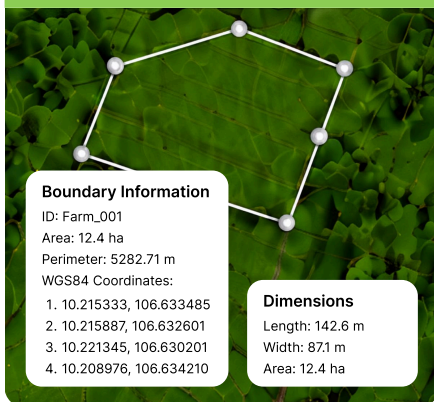


Efficient Management

Mapping data supports monitoring, operations, and more accurate decision-making.

Practical Applications

Farm Boundary Mapping



Accurately define area, perimeter, and boundary coordinates for farm planning and management.

Crop Mapping



Locate each crop with an ID/QR code, storing variety, planting date, condition, and care history.

Mechanization Support



Create movement routes for robots, sprayers, and autonomous equipment to operate more accurately.

Implementation Process

Install RTK Base

Set up the base station and stable power supply.

Data Transmission

Send survey data to the system via 4G/MQTT.

Manage & Utilize

Monitor and operate maps on app/web.

Survey with Rover

Move across the farm to collect RTK coordinates.

Process & Generate Maps

Generate 2D/3D maps, boundaries, crops, routes, and infrastructure.

UAV: MONSKY

Drone-based solution for agricultural surveying, monitoring, and traceability

MONSKY UAV is an AI-powered drone solution that captures multispectral imagery, analyzes crop health, and supports real-time decision-making.



Key Benefits

- Automated drone surveying: Cover large areas, plan flight routes, and collect growing-area data
- Individual crop monitoring: Track growth, canopy size, and crop health over time.
- AI-based pest detection: Identify pests, plant stress, and signs of nutrient deficiency early.
- Traceability & compliance: Digitize farm records to support certification and origin tracing.

Data Capabilities

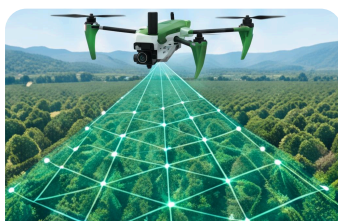
- Crop mapping
- Multispectral mapping
- Growth timeline
- Anomaly alerts

Practical Application



Visualize crop health in real time to support better farm decisions

Practical Applications



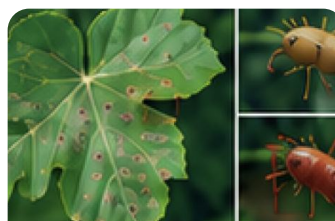
Growing Area Monitoring

Wide-area coverage, full monitoring, and time savings



Crop Health Mapping

Assess crop health using NDVI, NDRE, SAVI, and multispectral indexes



Pest & Stress Detection

AI detects pests, nutrient deficiency, and crop stress early



Traceability

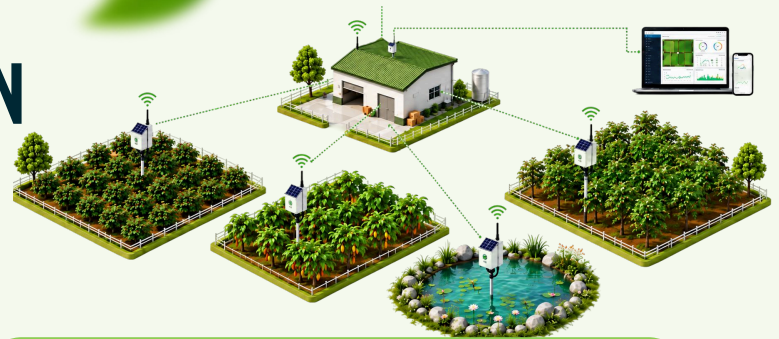
Digitize farm records to support certification and traceability

MONSKY UAV updates growing-area data in real time, detects crop stress and anomalies early, and supports AI-powered analysis for more accurate decisions. Digitized data also helps farms meet international standards and stay ready for traceability and export.

MULTI-SENSOR STATION

Soil, water, and weather monitoring solution for farms

One station supports up to 5 sensors, auto-detection, and mesh-node connection to the center for analysis and alerts by AgriBeacon AI Assistant.



Plug & Play

VY-SO · VY-WA · VY-AIR

AI Alerts

Supports a variety of sensors



VY-SO – Soil Nutrition (8): pH, EC, salinity, temperature, moisture, N, P, K.



VY-WA – Water Quality (5): pH, temperature, EC, TDS, salinity.



VY-AIR – Weather (8): temperature, humidity, wind speed & direction, pressure, rain, light, radiation.

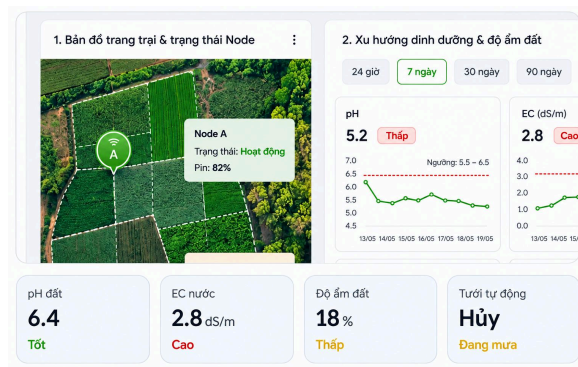


Flexible combination: choose sensors by location – orchard, pond, climate zone.

System Capabilities

- Plug & play: station auto-detects sensors
- Mesh – Node: stations connect to the nearest one, gateway aggregates and temporarily stores data
- History and trends by day / week / crop cycle
- Threshold and device status alerts
- Smart irrigation saves water: auto-start / stop based on real-time moisture and weather data
- Management alerts via app: on-site siren, manager alerts, and scheduled settings

Dashboard & AI Assistant



AgriBeacon AI Assistant integrates soil, water, and weather data for alerts, fertilizer suggestions, and timely auto-start / stop irrigation.

Deployment Models



Orchard – VY-SO

Pole + solar panel + control cabinet, VY-SO probe inserted from cabinet – monitors nutrition and moisture by zone.



Pond / Water Source – VY-WA

Station placed on shore, VY-WA probe lowered from cabinet into water – monitors pH, EC, TDS, and salinity.



Weather Station – VY-AIR

VY-AIR cluster mounted on pole and connect cabinet – rain, wind, and radiation serve as for irrigation and disease forecasting.



Mesh Node & Gateway

Stations form a mesh network. The gateway sends data to the center. Stable and easy to expand.

Small farm – standalone

Sends directly to the center; no gateway.

Large farm – Mesh ∞ stations

Stations connect to gateway via mesh, add gateways as needed.

Smart irrigation saves water

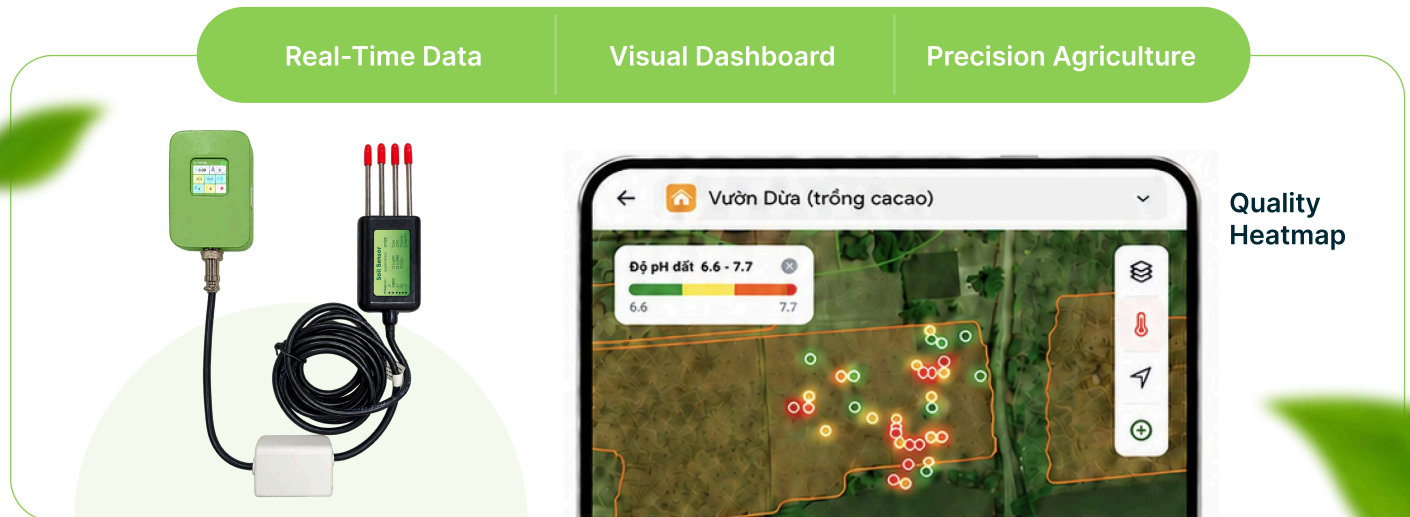
Auto-start / stop irrigation based on real-time moisture and weather data

Management alerts via app

On-site siren, manager alerts, and scheduled settings via app.

VY-SO SOIL NUTRIENT SENSOR

8-in-1 AIoT Solution



Why Monitor?



Nutrient Imbalance

Reduces yield and increases pest and disease risks.



Guesswork Fertilization

Leads to wasted costs and soil degradation.



Slow Lab Testing

Makes it difficult to track soil changes in real time.

Highlights

- Measures 8 indicators: N, P, K, pH, EC, moisture, temperature, salinity.
- BLE 5.0 sends real-time data to mobile devices
- Built-in OLED display + QR pairing for on-field readings
- Live dashboard & whole-farm soil quality heatmap
- Suitable for perennial crops: coffee, pepper, durian, cacao, fruit trees

Crop Applications

- Monitor seasonal soil health
- Detect early soil acidification via pH + EC
- Optimize fertilization by crop zone
- Reduce yield gaps between plots

Key Benefits

20%

Crop yield increase

24/7

Continuous monitoring

30s

Fast alert response

2X

Payback in the first season

VY-WA WATER QUALITY SENSOR

5-in-1 AIoT solution for irrigation water quality monitoring



Real-time data



Dashboard & alerts



4G/LTE



IP68 rated



Why It Matters

- Poor irrigation water quality can increase plant mortality and limit growth by up to **40%**
- Contaminated irrigation water can reduce crop yield by up to **25%**
- Manual testing is slow, lacks continuous data, and makes early alerts difficult

Highlights

- Measures 5 indicators at once: pH, DO, ORP, temperature, TDS
- 4G/LTE connection with continuous data transmission
- Web/app dashboard with real-time alerts
- IP68 waterproof, built for outdoor use
- Solar / DC 12V power supply

Practical Application



5 Key Parameters



pH
0 – 14 pH
Range: ±0.1 pH



TEMP
-10 – 60°C
Range: ±0.3°C



DO
0 – 20 mg/L
Range: ±0.3 mg/L



ORP
±2,000 mV
Range: ±5 mV



TDS
0 – 10,000 ppm
Range: ±2%

From Metrics to Farming Decisions



pH
Controls acidity/alkalinity for better nutrient absorption.



TEMP
Tracks water temperature changes.



DO
Measures dissolved oxygen to assess water quality and aeration.



ORP
Evaluates water cleanliness and organic/microbial pollution risks.



TDS
Controls salts, dissolved minerals, and root-related risks.

Typical Applications



Healthy aquaculture



Smart irrigation



Risk prevention



Data-driven decisions



Monitor water quality directly in ponds or irrigation water sources

ROBOTS: DUBIRO

Autonomous / Semi-Autonomous Spraying Robot
Solution for Modern Farms

Operate remotely and automate the spraying process to ensure safety, consistent coverage, and complete traceability data after every run.



Remote
Operation



RTK
GNSS



4G / LTE



360°
Panoramic
Camera



Highlights



Reduces direct contact with
chemicals



Consistent, stable route-
based spraying



Flexible operation: remote
control or semi-autonomous
mode



Real-time monitoring via 360°
camera video stream



Records route history, flow
rate, and spraying time



Multiple nozzle options for
different crop types

Control & Monitor Live from Anywhere



4G / LTE

Live Video

Real-Time Monitoring

Benefits



Safer



More Consistent



Easier to Control

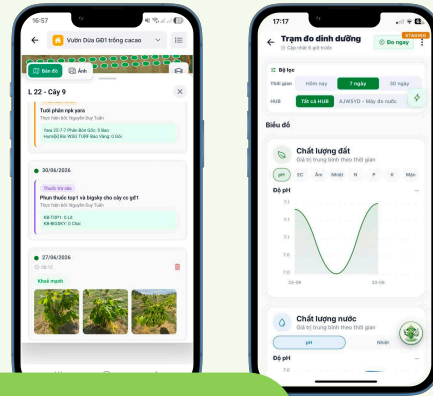


Traceable Data

SUTAGROW

Farm operations management platform
Daily execution. Traceability down to every tree.

Plan, assign tasks, log activities, and track evidence on one platform

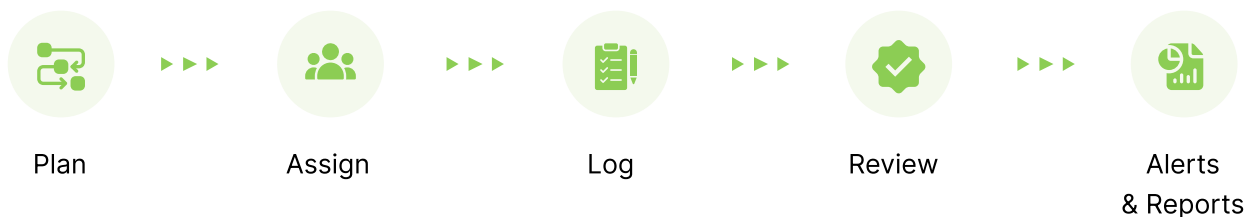


In the Field

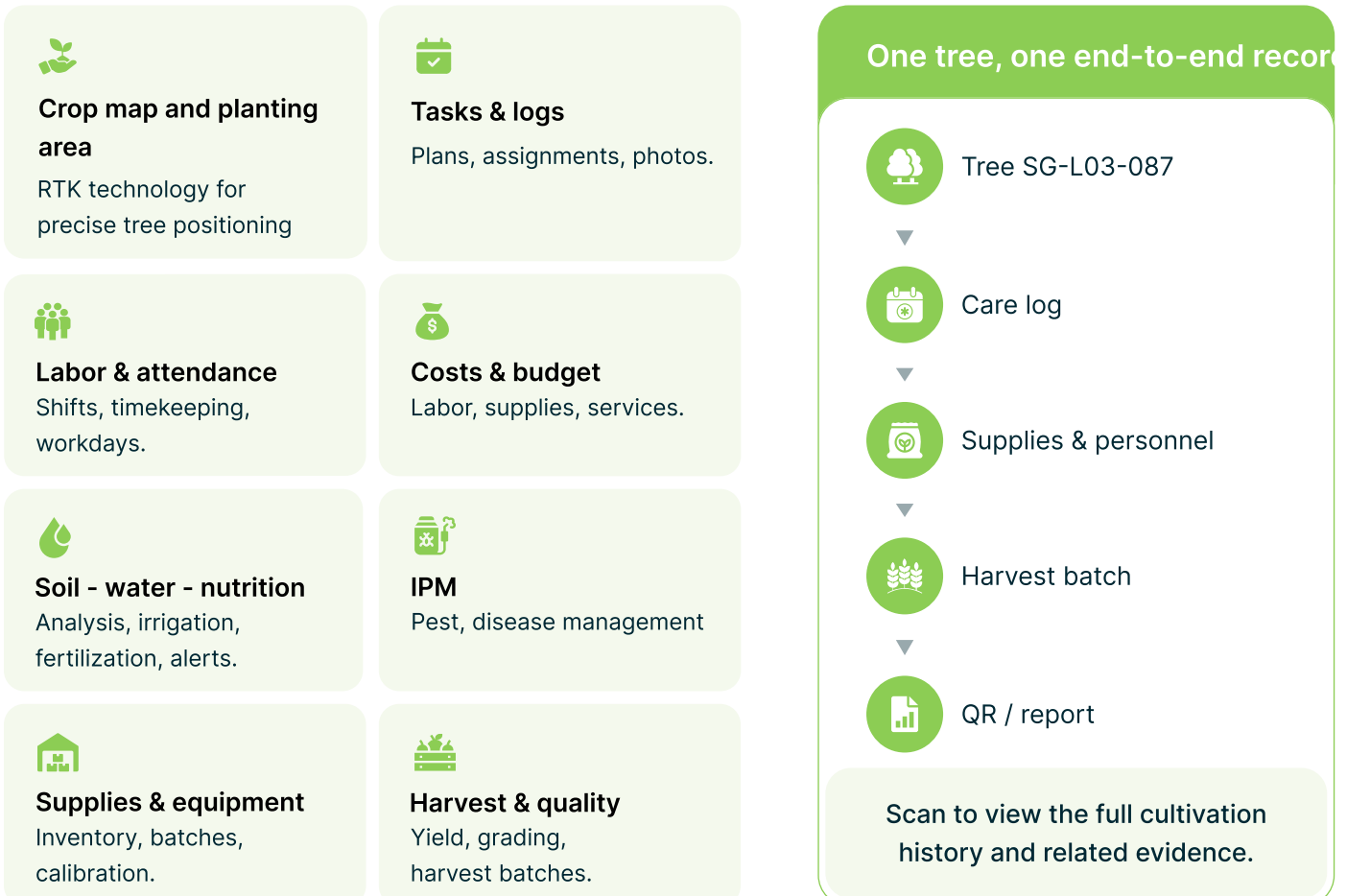
By Plot / Area / Tree

Web - iOS - Android

A day of operations on Sutagrow

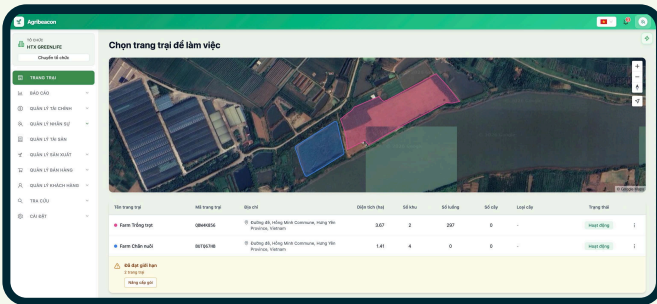


Comprehensive operational management



MANAGE MULTIPLE FARMS ON A SINGLE OPERATIONS MAP

Track multiple farms, growing areas, and land plots on one unified operations platform with accurate field-based location data.



Multi-Farm Management

Monitor multiple farms, growing areas, and crop seasons from one centralized dashboard.

AgriBeacon standardizes operational data by location, helping businesses manage farm networks clearly, consistently, and at scale.

Plot Zoning

Crop Cycle

Cross-Location KPIs

GIS / RTK Map

Digitize farm and land-plot boundaries using accurate positioning data.

AgriBeacon connects maps with sensors, drones, and field operations to manage growing areas by zone and support traceability.



AI FARM ASSISTANT

Understand each role. Guide the right tasks.

Connect operational, field, and agronomic data so everyone can see what they need to know, what to do, and what to review.



Each role, a tailored experience



Farm Owner

Performance, cost, and risk



Operations Manager

Prioritize, coordinate, and control



Agronomist / Field Officer

Crop care, nutrition, and pests



Field Worker

Receive tasks, complete them, and log them

Each role, a tailored experience



See the full operational picture

Track tasks, labor, workdays, fertilizers, supplies, costs, and field data.



Agronomy explained clearly

Explain farming techniques so new team members can do the right thing.



Identify what needs review

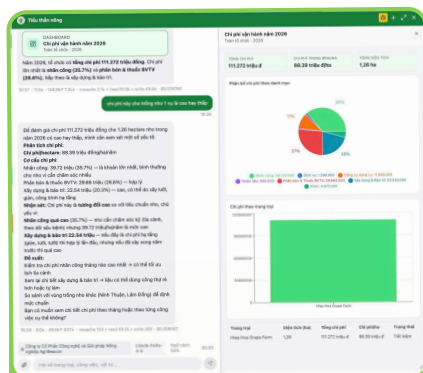
Detect unusual figures, missing logs, and operational gaps.



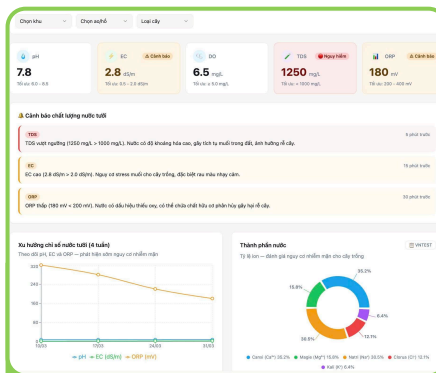
Smart data entry with AI

Turn voice and text into tasks and instant suggestions — free you from searching and filling data.

More than answers, we take action with you



Analyze operations & costs

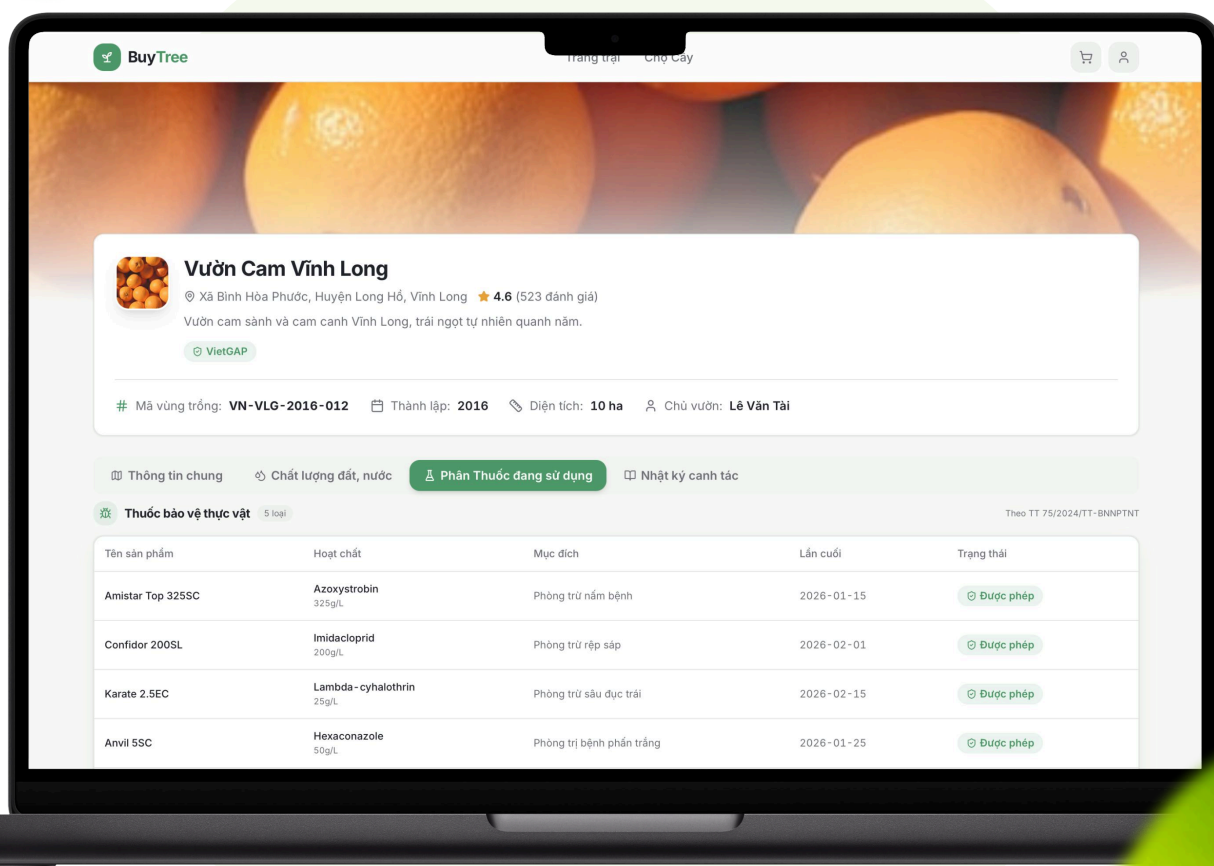


Field alerts from field data

Conversations become prefilled suggestions

SUTAMARKET FARM LOOKUP & TREE PURCHASE PLATFORM

SutaMarket connects farm profiles, cultivation data, and market demand, helping farms showcase real value through verified data.



Individual Tree Ownership

Buyers can look up tree information, growing areas, and care history through clear, transparent records.



Verified Quality

Cultivation data, input records, field images, and operation history help demonstrate farm quality with confidence.

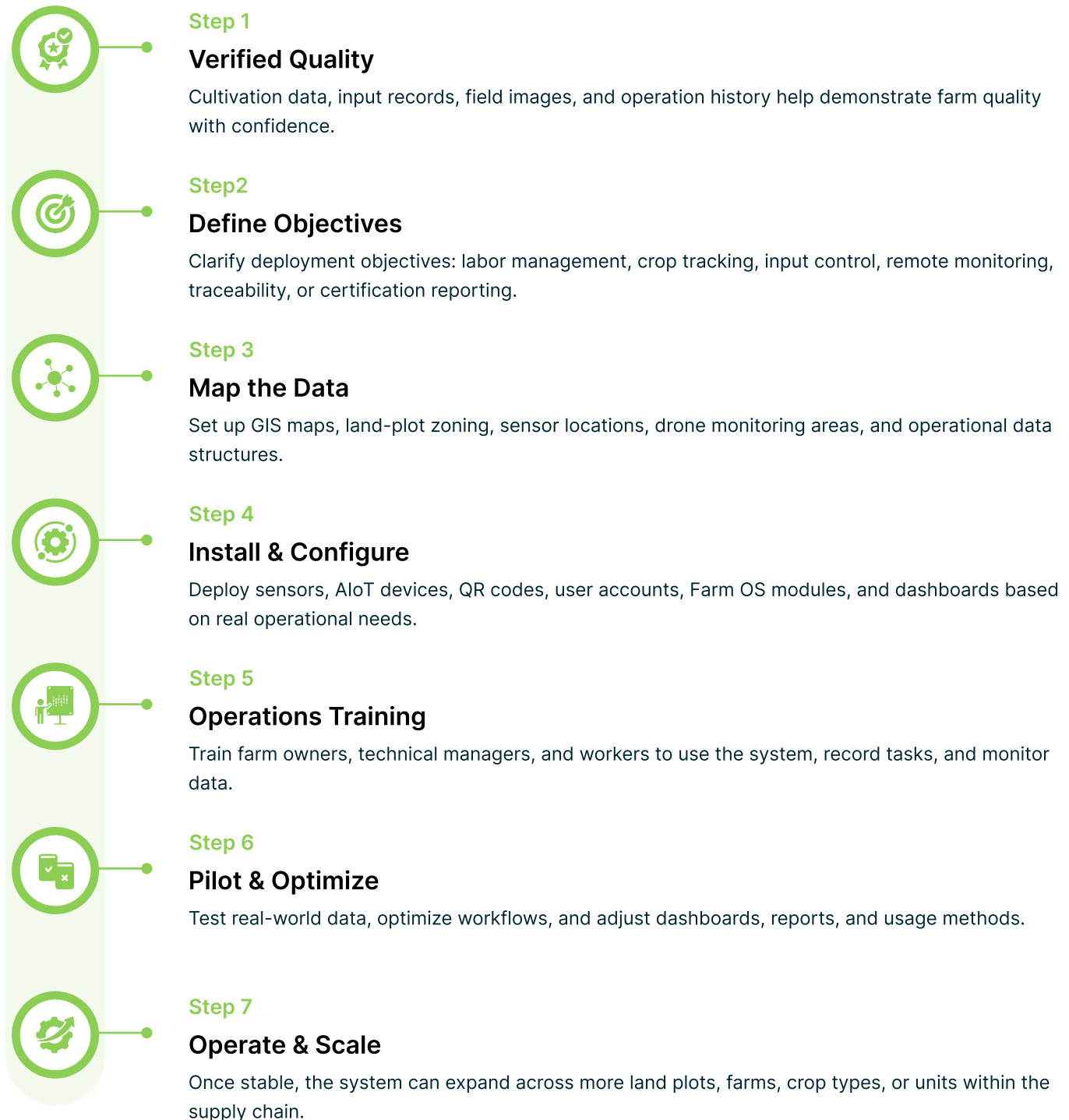


Transparent Payments

SutaMarket connects farms, buyers, and purchasing partners through a clear data platform, reducing reliance on manual commitments.

FIELD SURVEY TO REAL-WORLD OPERATIONS

AgriBeacon provides more than standalone software or individual devices. We deploy an operations system tailored to the real conditions of each farm, cooperative, or enterprise.



AgriBeacon helps customers move from field surveys to systematic data-driven operations, reducing tool fragmentation and building a scalable management foundation.

AIOT DEVICES

AgriBeacon provides an AloT device ecosystem that collects field data from soil, water, maps, drones, and robots, forming the foundation for farm monitoring, AI analytics, and precision operations.



RTK base

Accurate coordinate reference base



Agricultural Monitoring Drone (UAV)

Monitor and observe individual crops



VY-NO Multi-Sensor Monitoring Station

Supports multiple sensors with periodic data and location updates.



8-Parameter Soil Nutrient Sensor

(pH, EC, N, P, K, Moisture, Temperature).



5-Parameter Water Quality Sensor

(pH, EC, TDS, DO, ORP).



Autonomous Multi-Task Robot

Autonomous spraying and grass cutting



RTK Rover

Map crops and farms using RTK technology



Seamlessly connect source-level data into a unified management system, enabling AI to turn raw data into breakthrough growth solutions.

READY TO TAKE YOUR FARM TO THE NEXT LEVEL?

From in-field sensors to compliance proof for buyers, AgriBeacon helps farms operate smarter, become more sustainable, and connect with better markets every day.





**SMART SENSORS,
ACCURATE DATA,
SUSTAINABLE GROWTH**

Download the SutaGrow mobile app



Google Play



Apple Store

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